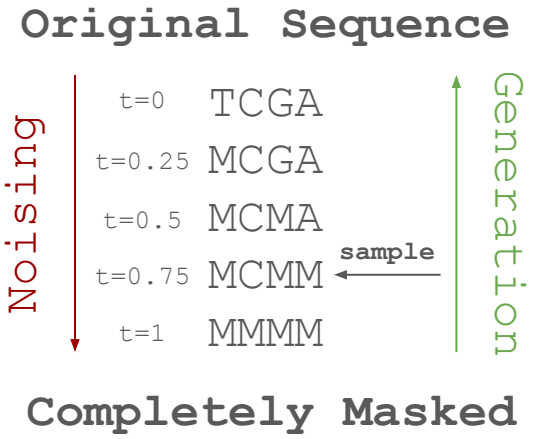
Diffusion Sampling via Information Theory

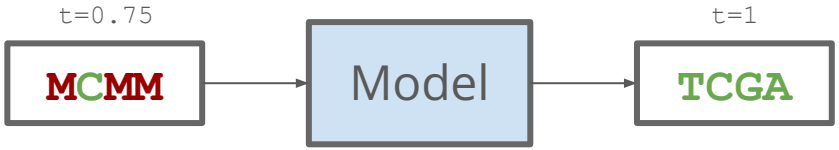
Jai Sharma, Yifan Wang, Bryan Li

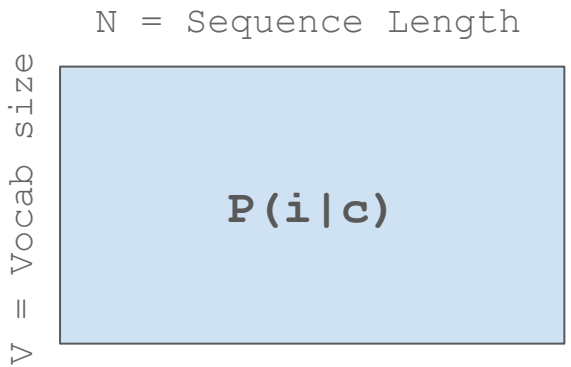
Introduction to Masked Diffusion Models

- MDMs estimate joint structure in systems where variables influence one another.
- Examples: generation of text, DNA, proteins, graphs
- Used when dependencies are complex and cannot be captured by simple autoregressive orderings.
- Used where global consistency is more important than temporal order
- Can be generated in any order, and in parallel!

Training Masked Diffusion Model







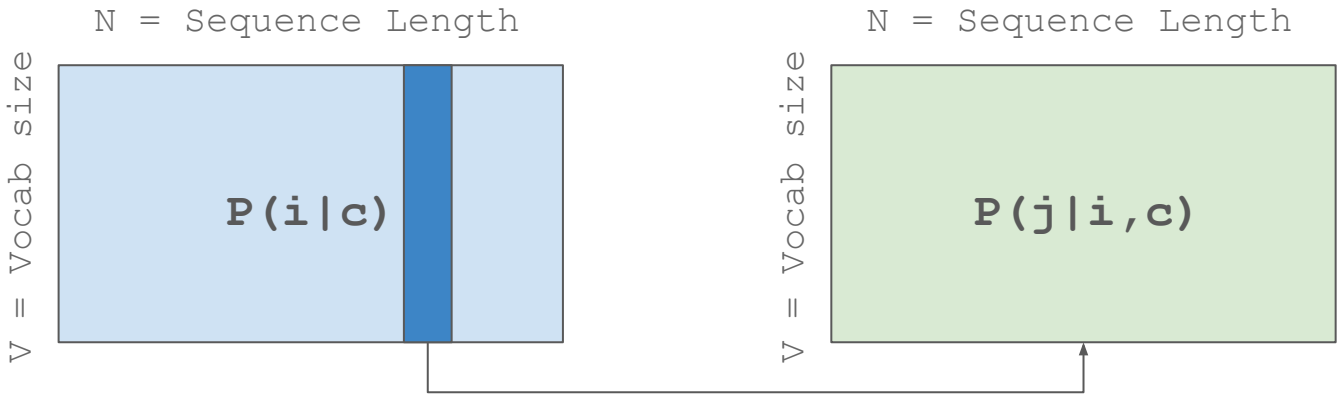
Case Study: Sudoku

Introduction to Masked Diffusion Models

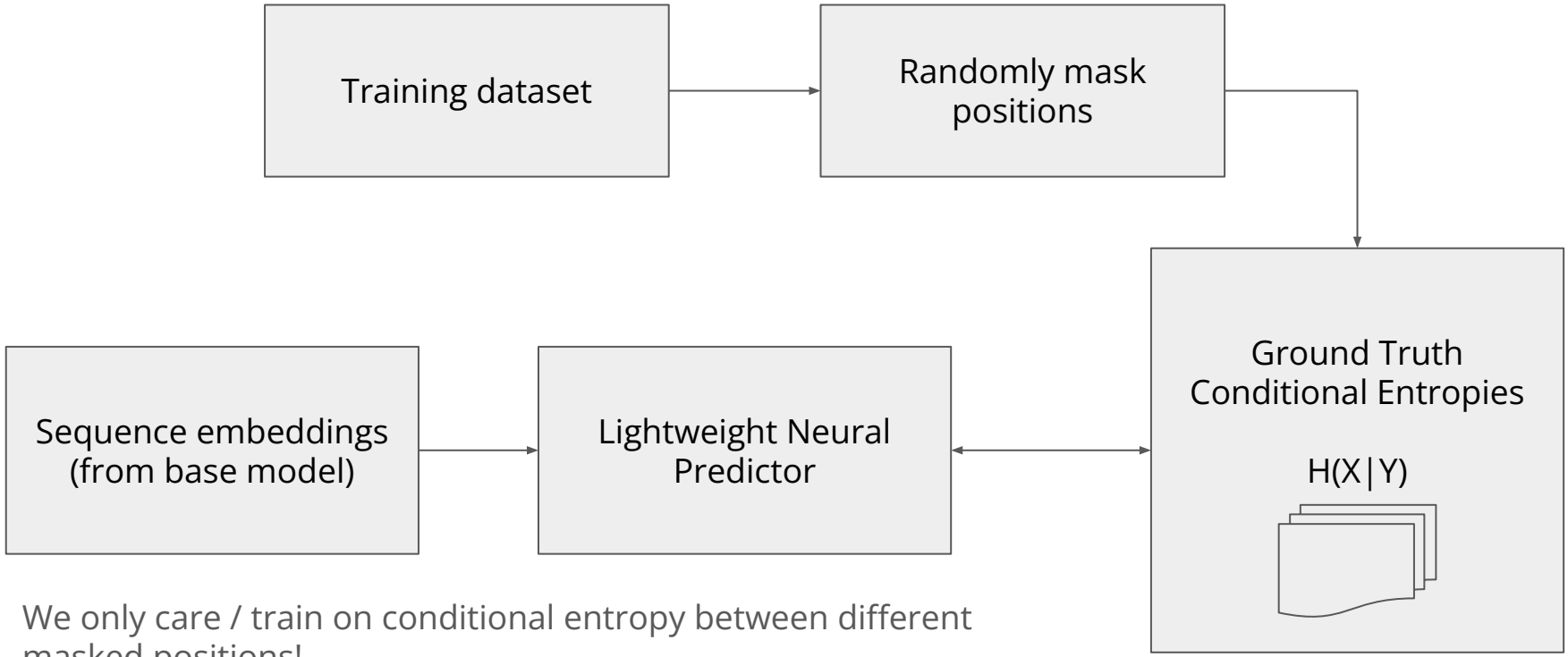
- Problem: Inference is slow!
- Our goal: speed up inference by parallelizing unmasking process while maintaining performance

Calculating Ground Truth Mutual Information

```
for v in VOCAB:
    x[i] = v
    pass seq into
model
```

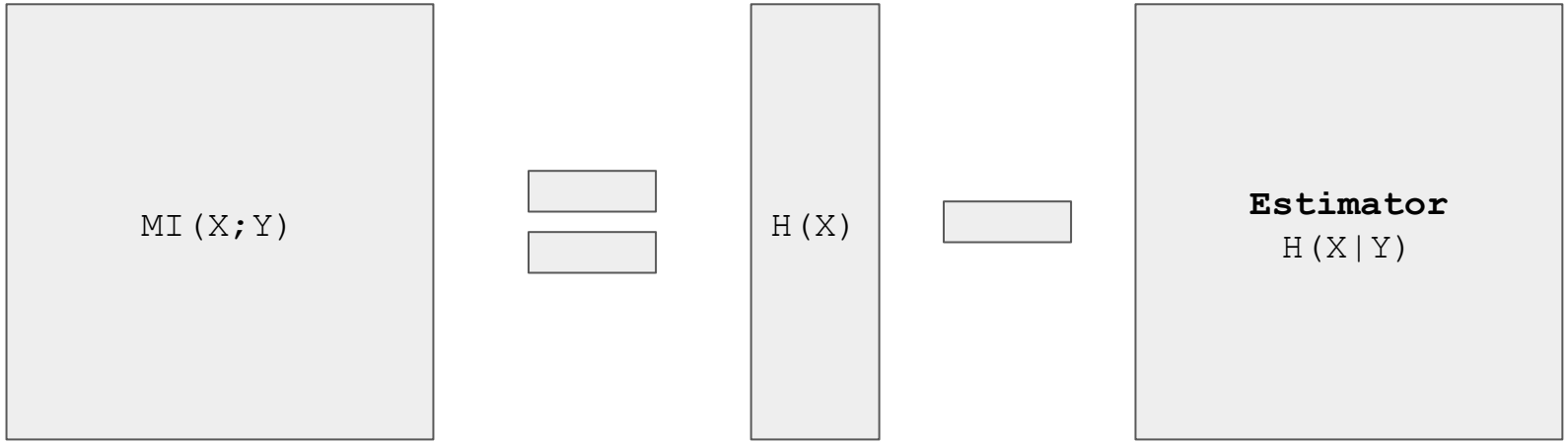


Neural Estimator for Mutual Information



We only care / train on conditional entropy between different masked positions!

Recovering the mutual information

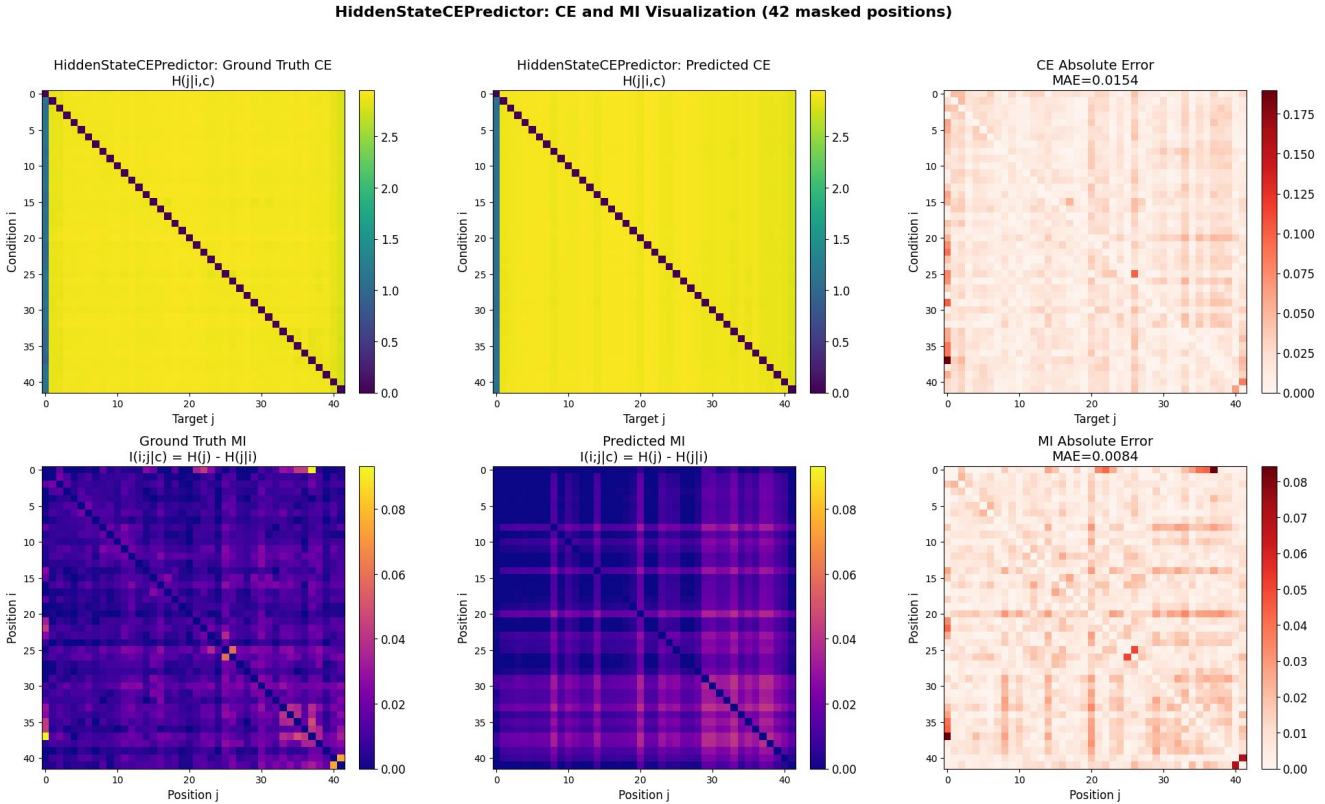


Note: as a precaution, we symmetrize the mutual information matrix, since the model's outputs (and thus entropies) may be noisy

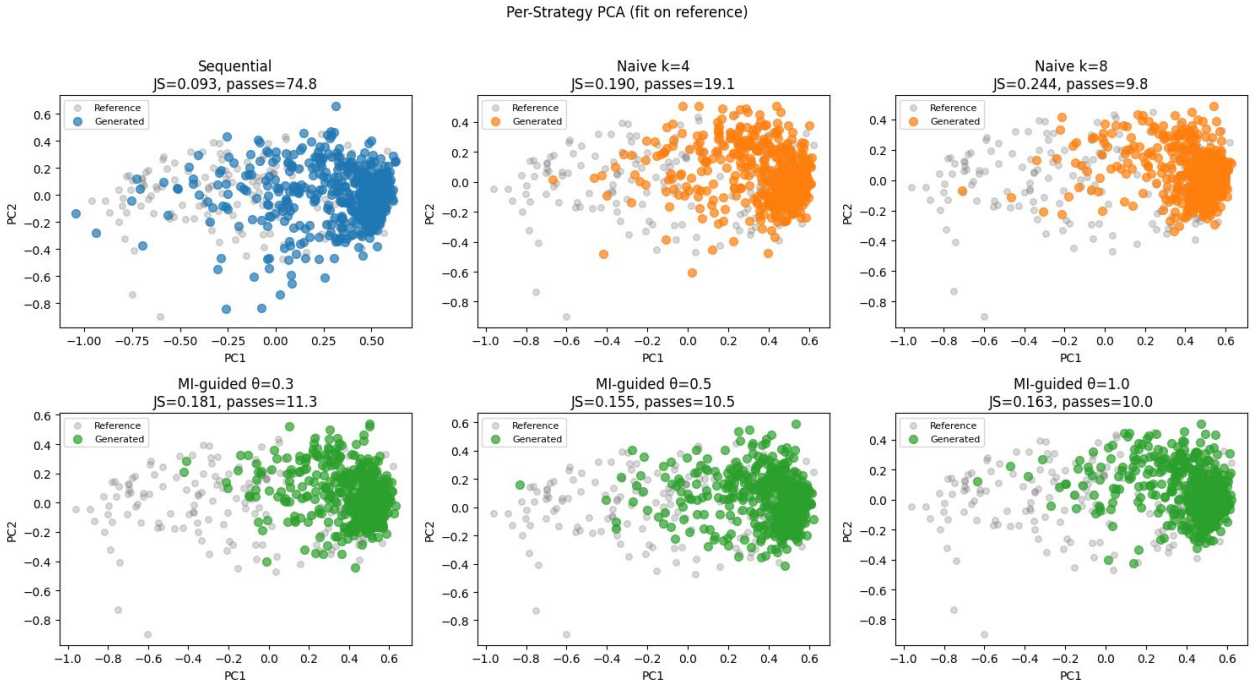
Applications: Sampling using Mutual Information

1. For each masked position, compute entropy/confidence
2. Estimate pairwise dependency using mutual information
3. Define entropy_weight = 1 - entropy / max(entropies)
4. Define gain(i) = MI(i; rest_of_masked) (mean over remaining masked)
5. Define redundancy(i,S) = average MI(i;s) over already selected S.
6. Score(i) = entropy_weight * gain(i) – λ·redundancy(i,S).
7. Use a greedy approach to pick up positions.

Applications: Sampling using Mutual Information



Applications: Sampling using Mutual Information

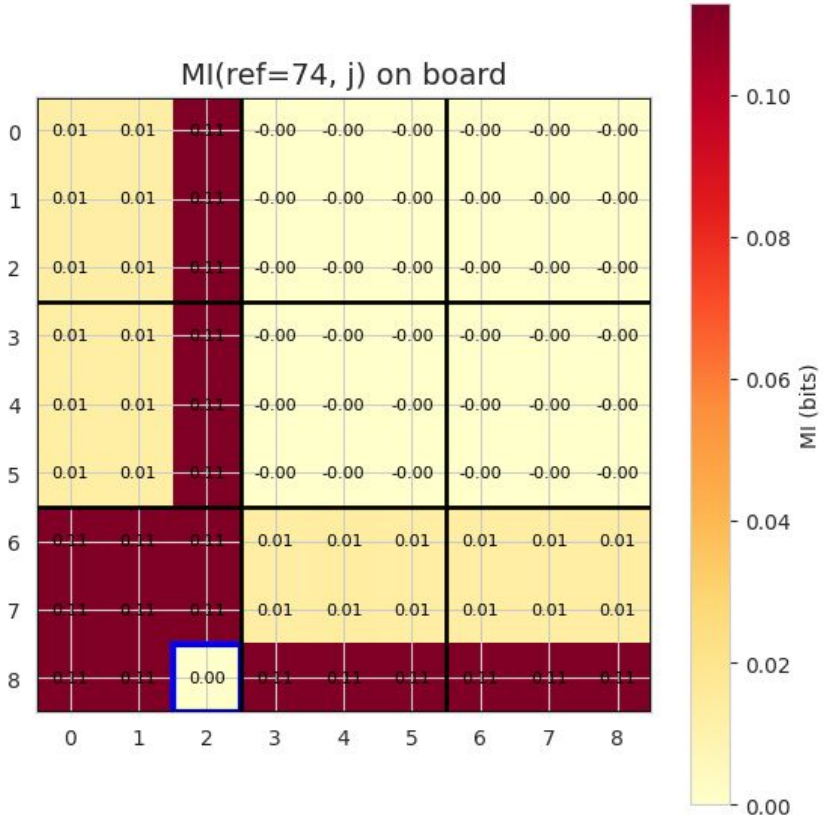



```
Loading cached results from unconditional_res
Loaded results for 6 strategies
  Sequential: JS=0.0925, passes=74.8
  Naive k=4: JS=0.1897, passes=19.1
  Naive k=8: JS=0.2439, passes=9.8
  MI-guided θ=0.3: JS=0.1813, passes=11.3
  MI-guided θ=0.5: JS=0.1546, passes=10.5
  MI-guided θ=1.0: JS=0.1632, passes=10.0
```

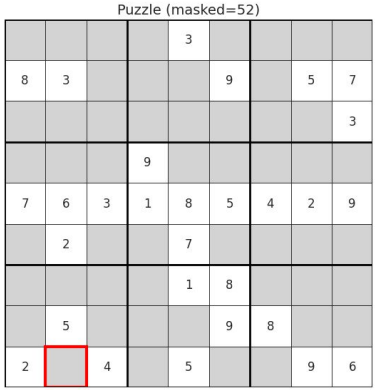
Applications: Mutual Information Interpretability

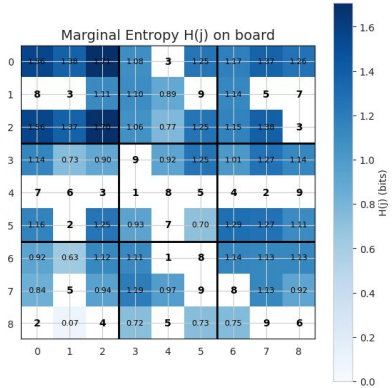
Notice the mutual informations are 0.11 nats!

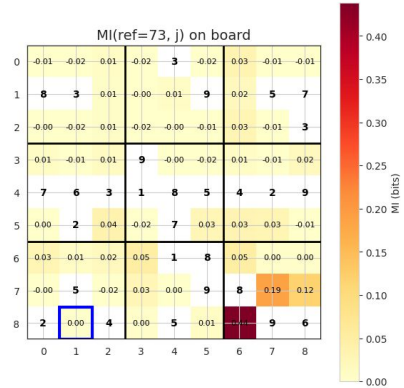
P(same col/row/box | reference) goes from U[9] to U[8]
ln(9) - ln(8) = 0.11 nats

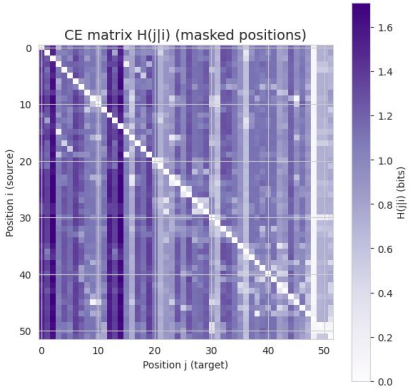


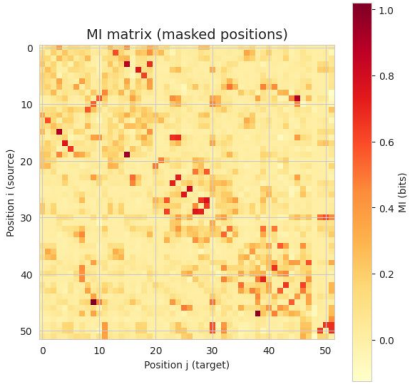
Applications: Mutual Information Interpretability

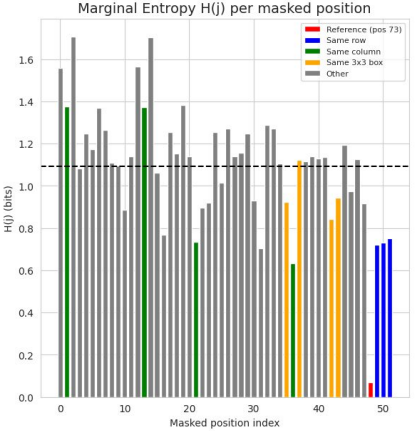












Applications: Mutual Information Interpretability

